

Liquidity and Asset Prices

Learning Objectives:

- What is the illiquidity premium?
- Role of the holding horizon and liquidity clienteles
- Determinants of the bid-ask spread and illiquidity premia
- Search frictions, execution delay, and illiquidity premia in OTC markets
- Liquidity risk and illiquidity premia

10.1 Introduction

A recurrent theme of this book is that the various costs of liquidity provision (attributable to adverse selection, inventory holding, and order processing) create a wedge between transaction prices and the fundamental values of assets. This wedge is a measure of market illiquidity. But we have not yet considered that illiquidity may affect the asset's value.

The main reason for this is that transaction costs reduce the return to investors, just as a tax on capital gains does. Hence, insofar as they expect to pay transaction costs in the future, investors are willing to pay less for less liquid assets, so asset returns contain an illiquidity premium in addition to a risk premium. As investors care about asset returns net of expected trading costs, the illiquidity premium is determined both by the bid-ask spread and their trading frequency, as the latter determines how often they “pay the bid-ask spread.”

Section 10.2 makes this point focusing on the simple case where both the bid-ask spread and investors' trading frequency are taken to be exogenous, and shows that an asset's illiquidity premium depends not only on its bid-ask spread but also on the horizon over which investors wish to hold the asset: the greater the holding period, the lower the impact of the bid-ask spread on investors' valuation of the asset, because they pay the bid-ask spread less frequently. Insofar as investors have different desired holding periods, they will require different illiquidity premia from assets, and this can generate “investor clienteles”: investors that tend to trade more frequently will tend to prefer assets with lower bid-ask spreads while those that trade infrequently will specialize in

illiquid assets. As shown in Section 10.2.2, in the presence of such clienteles, the illiquidity premium should increase less than proportionately with the bid-ask spread.

Since the illiquidity premium reflects investors' expected trading costs, it is affected by all the determinants of the bid-ask spread examined in Chapter 3: the higher are adverse selection, inventory holding costs, and order processing costs, the higher the illiquidity premium, hence the lower the market price and the higher the expected return that an asset will command in equilibrium. Section 10.3 illustrates this point with reference to the adverse selection component of the bid-ask spread, and presents empirical evidence indicating that assets with greater probability of informed trading feature a higher illiquidity premium.

In centralized markets investors can execute their trades without delay. In contrast, in decentralized markets, such as OTC ones, order execution takes time because trading requires searching for a counterpart. This search generates delays in execution and, insofar as these delays are costly, they will also contribute to the illiquidity premium, in addition to the bid-ask spread that investors pay when trading. In addition, the delays caused by search frictions confer market power on dealers, and therefore increase the bid-ask spread charged to investors, thereby contributing to the illiquidity premium also via this channel, as explained in Section 10.4.

Risk-averse investors can be expected to care not only about expected trading costs but also about their randomness: as liquidity varies over time in ways that cannot be fully anticipated, its fluctuations may add "liquidity risk" to the fundamental risk of assets, such as default risk and unknown future dividend streams. As usual in asset pricing theory, if investors are sensitive to risk, they will require compensation for liquidity risk as well, unless it can be diversified. Liquidity risk and how it affects illiquidity premia is discussed in Section 10.5.

10.2 The Illiquidity Premium

This section presents a simple framework drawn from Amihud and Mendelson (1986), which illustrates the effect of liquidity on asset prices and required returns and shows how this relationship is affected by the investors' time horizon, with empirical evidence on the relationship.

10.2.1 Illiquidity Premia and Investors' Holding Period

To illustrate how the level of liquidity affects the prices of asset, consider a simple example. U.S. Treasury notes and bills are bonds of different initial maturity. They have exactly the same default risk but are traded on distinct

markets. At issue, notes have longer maturities (two to ten years) than bills (twelve months or less). But at any given point in time there will coexist bills and notes with identical residual maturity of less than six months, with a single payout remaining at maturity. As those securities are identical in risk and payout timing, standard arbitrage arguments would imply that in a frictionless world they should have the same price per dollar of payout. However, Amihud, and Mendelson (1991) (and many subsequent papers) find that notes typically trade at a substantial discount relative to identical bills; on average, at the time of their study, the annualized yield on notes exceeded that on bills by 43 basis points, in a sample with an average maturity of about 95 days.

A natural explanation for this significant premium is that the markets for bills and notes are not equally liquid. Amihud and Mendelson (1991) report that the bid-ask spread and brokerage fees on notes are greater than that on bills. Hence notes must offer extra compensation to attract investors.

To see this, consider the following simple model. An investor buys a security that he plans to sell after h periods; during this time the security does not pay dividends or interest. The market is illiquid, and the proportional bid-ask spread at date t is s_t , so that the ask and bid prices at that time are given by:

$$a_t = m_t \left(1 + \frac{s_t}{2}\right), \quad (10.1)$$

$$b_t = m_t \left(1 - \frac{s_t}{2}\right). \quad (10.2)$$

Henceforth, we assume that the midquote is equal to the fundamental value of the security, that is, $m_t = \mu_t$. For brevity, we call s_t a bid-ask spread, but it should be interpreted more broadly as a measure of trading cost, which includes, for instance, the price impact of large transactions and brokers' commissions. Suppose that investors require a return of r per period on the security, given its risk characteristics. For example, for a riskless security like a Treasury bill, the required return is simply the risk-free rate. If the market were perfectly liquid (i.e., $s_t = 0$ at any time) then μ_t would grow at rate r on average.

Under these assumptions, the maximum price that the investor is willing to pay is given by the standard discounted cash flow model:

$$a_t = \frac{b_{t+h}}{(1+r)^h}, \quad (10.3)$$

that is, using (10.1) and (10.2):

$$\underbrace{\mu_t \left(1 + \frac{s_t}{2}\right)}_{a_t} = \underbrace{\mu_{t+h} \left(1 - \frac{s_{t+h}}{2}\right)}_{b_{t+h}} \frac{1}{(1+r)^h}.$$

This equation can be used to express the current value of the asset as its discounted future value adjusted for current and future transaction costs:

$$\mu_t = \mu_{t+h} \times \frac{1}{(1+r)^h} \times \frac{1 - \frac{s_{t+h}}{2}}{1 + \frac{s_t}{2}}. \quad (10.4)$$

The last term is a measure of illiquidity because it decreases in both s_t and s_{t+h} . The fundamental value at date t is determined by illiquidity: the greater the current or future spread, the higher the transaction costs for investors and the lower the value of the asset to them. This is consistent with our example, in which the more liquid Treasury bills traded at a premium over notes, notwithstanding their identical payoffs at maturity.

Empirical researchers often find it easier to test asset pricing models by investigating the properties of returns rather than prices. Therefore, we use equation (10.4) to derive a relationship between gross returns and bid-ask spreads:

$$\frac{\mu_{t+h}}{\mu_t} = (1+r)^h \times \frac{1 + \frac{s_t}{2}}{1 - \frac{s_{t+h}}{2}},$$

and thus R , the gross return per period over the holding horizon h , is defined by the following expression:

$$(1+R)^h = (1+r)^h \times \frac{1 + \frac{s_t}{2}}{1 - \frac{s_{t+h}}{2}}. \quad (10.5)$$

The gross return R is the average per-period percentage change in the security's fundamental value required to induce the investor to hold the security. As the last term on the right-hand side of the equation exceeds 1, R is greater than the net return r : the difference compensates investors for transaction costs. This explains why the yield on notes can be greater than the yield on bills.

The gross return in (10.5) also depends on the investors' holding period h . It is easiest to see the effect of this factor if the percentage spread s_t is assumed constant, so that $s_t = s_{t+h} = s$. Using this assumption in equation (10.5), the gross return R is given by the following approximation (for r and s small):¹

¹ A way to derive this approximation is to take logarithms of both sides in equation (10.5) and recall that $\ln(1+x) \simeq x$, for x small.

$$R \simeq r + s/h. \quad (10.6)$$

This expression shows that R is increasing in the spread s at a rate that is inversely proportional to the holding period h . The reason is that an investor who holds the security incurs the transaction cost only once every h periods, and thus the per-period cost is reduced.

The foregoing analysis may seem restrictive, in that it assumes the asset to be safe. But this assumption is not indispensable: if μ_t (the fundamental value) is uncertain, the discount rate should be the rate of return required by investors, given the risk of the security. According to the capital asset pricing model (CAPM), this risk adjustment is determined by adding to the risk-free rate r a premium that is proportional to its “beta,” that is, the covariance between the return on the stock and that on the market portfolio, divided by the variance of the latter. Specifically, the required rate of return on asset j should be:

$$E(r_j) = r + \beta_j[E(r_M) - r], \quad (10.7)$$

where $E(\cdot)$ is the expectation operator, r_M is the uncertain return of the market portfolio, and $\beta_j \equiv \text{cov}(r_j, r_M)/\text{var}(r_M)$. This expression indicates that the risk premium is the product of the contribution of asset j to the risk of a well-diversified portfolio (β_j) multiplied by the risk premium on the market portfolio ($E(r_M) - r$). Hence, the β of a security captures its specific risk, which cannot be suppressed by holding it in combination with many others. For this reason, this parameter is often called the security’s systematic risk.

For asset j , equation (10.5) becomes:

$$[1 + E(R_j)]^h = [1 + E(r_j)]^h \times \frac{1 + \frac{s_{jt}}{2}}{1 - \frac{s_{jt+h}}{2}}, \quad (10.8)$$

where R_j is the gross return, and s_{jt} the bid-ask spread at date t . Substituting $E(r_j)$ from equation (10.7) into (10.8) and making the same approximation as before, we obtain:

$$E(R_j) = r + s_j/h + \beta_j[E(r_M) - r]. \quad (10.9)$$

This expression shows that the required gross rate of return on asset j is equal to the risk-free rate plus the illiquidity premium s_j/h plus the risk premium $\beta_j[E(r_M) - r]$. Hence, the CAPM relationship (10.7), which applies to net returns (because that is what investors care about), must be suitably modified if it is estimated using gross returns (i.e., in our setting, returns based on mid-quotes

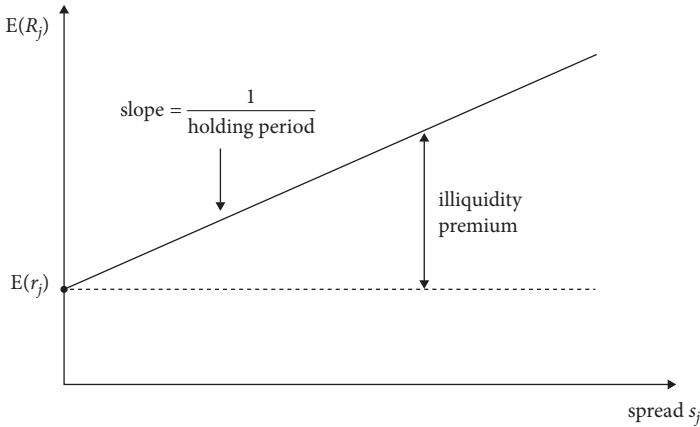


Figure 10.1 Expected return and bid-ask spread.

instead of transaction prices). Figure 10.1 depicts the relationship between the expected gross return on asset j and the spread, where the intercept is the expected net return, $E(r_j) = r + \beta_j[E(r_M) - r]$, and the slope is $\frac{1}{h}$, the inverse of the holding period.

Thus gross expected returns have two determinants: (i) the classical determinants of required asset returns, such as the beta in the CAPM, and (ii) an additional premium that compensates for transaction cost. We refer to this premium s_j/h as the illiquidity premium. If all investors have the same holding period h , then the illiquidity premia are approximately linearly related to spreads, with slope $1/h$, as depicted in Figure 10.1. If equation (10.9) holds in reality, the inverse of the coefficient of the spread in a regression for the gross rate of return should provide an estimate of the holding period of the representative investor.

Another assumption in the foregoing analysis is that the future spread is known with certainty at the initial date. But in reality the spread may change unpredictably, so that the riskiness of the liquidation price will be determined both by fluctuations of the fundamental value μ_{t+h} (due to news about future cash flows, for example) and by fluctuations in the spread s_{t+h} . The latter generate what is known as “liquidity risk.” If the liquidity risk of a security is idiosyncratic—that is, uncorrelated with market returns—then the analysis is unchanged, except that the actual spread s_{t+h} must be replaced by its expected value $E(s_{t+h})$. Investors will not require any compensation for liquidity risk, since they can diversify it away.²

² According to the CAPM, an asset risk premium is determined by the co-variation of its net return with the net return of the market portfolio. If trading costs are uncorrelated across securities,

In reality, however, future liquidity tends to worsen precisely at times of low market returns (i.e., when μ_{t+h} is low) and high fundamental volatility. This means that liquidity risk amplifies the systematic fundamental risk of securities—it increases their beta. In equilibrium, this prompts investors to demand an additional compensation for liquidity risk, as analyzed in more detail in Section 10.5.

Finally, the analysis can be extended to the case in which the security pays cash flows (coupons or dividends) before the liquidation date $t + h$. These cash flows do not affect the basic insights developed so far. However they reduce the magnitude of the liquidity premium because investors receive part of the return directly via these cash flows, without having to liquidate the security (see exercise 1).

10.2.2 Clientele Effects

So far, we have implicitly assumed that all investors have the same holding period. But in practice different investors typically have different holding periods. For instance, pension funds tend to hold securities for longer periods than mutual funds. The cost of trading weighs more heavily on investors who trade frequently, so one would expect them to prefer highly liquid stocks, while low-turnover investors should be more willing to invest in illiquid securities. Amihud and Mendelson (1986) show that this “clienteles effect” implies that equilibrium gross returns should be a concave rather than a linear function of the spread.

To see this, consider a simple example with just two securities, a liquid zero-coupon bond with spread s_1 and a relatively illiquid one with spread $s_2 > s_1$, and two types of investors, one with short holding period h_1 and another with long holding period $h_2 > h_1$. Consider a situation in which the first group buys only the liquid bond at the gross rate R_1 and the second buys the illiquid bond at the rate R_2 .

This situation is an equilibrium if each of the two clienteles has no incentive to switch out of its candidate “preferred habitat” (in the wording of Modigliani and Sutch 1966). This is the case for the first group of investors, if they earn at least as large a net return r_1 by investing in the liquid asset as by investing in the illiquid one:

$$R_1 - s_1/h_1 \geq R_2 - s_2/h_1.$$

they do not contribute to the cross-sectional covariations in net returns and therefore do not affect the risk premium required to hold a security.

Conversely, the second group of investors will not want to switch to the liquid security if their net return on the illiquid security, r_2 , is at least equal to the return per period on the liquid security:

$$R_2 - s_2/h_2 \geq R_1 - s_1/h_2.$$

These two conditions, taken together, imply the inequality:

$$\frac{1}{h_2} \leq \frac{R_2 - R_1}{s_2 - s_1} \leq \frac{1}{h_1}. \quad (10.10)$$

This condition implies that investor 1 is in his preferred habitat purchasing security 1, while investor 2 prefers security 2. Graphically, the middle expression in inequality (10.10) is the slope of the segment BC that connects the equilibrium return-spread pairs for the two assets in Figure 10.2. The condition requires it to lie between the indifference curve of clienteles 1 and 2: the indifference curve of clientele i depicts the gross return-spread combinations that yield a given net return r_i to clientele i . Thus, the indifference curve of clientele i has slope $1/h_i$, for $i = 1, 2$. In the figure, the short-horizon clientele 1 prefers point B to point C (i.e., prefers security 1 to security 2). It also prefers point B to point A, which corresponds to a zero-spread asset. Conversely, clientele 2 is best off at point C, and therefore will hold security 2. Group 2 cares less about liquidity because it has a longer holding period, and so finds the extra return associated with a high spread s_2 sufficiently attractive to prefer C over B, while group 1 cares more about liquidity and therefore prefers B over C.

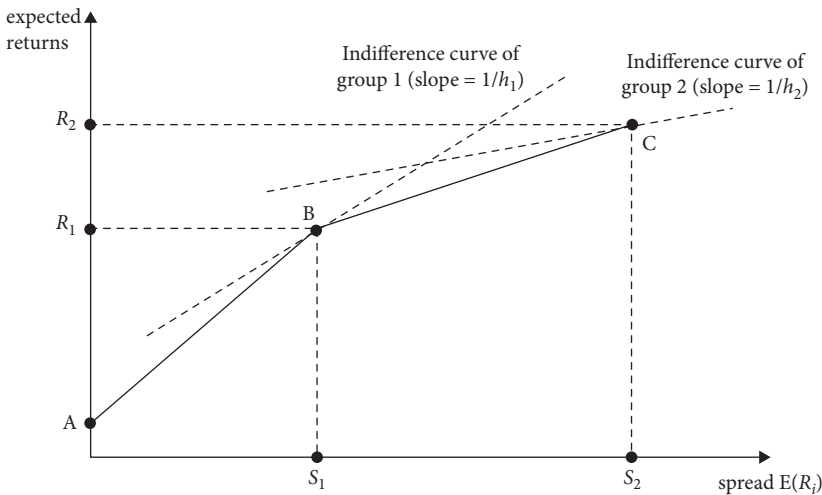


Figure 10.2 Expected return and spread with two clienteles.

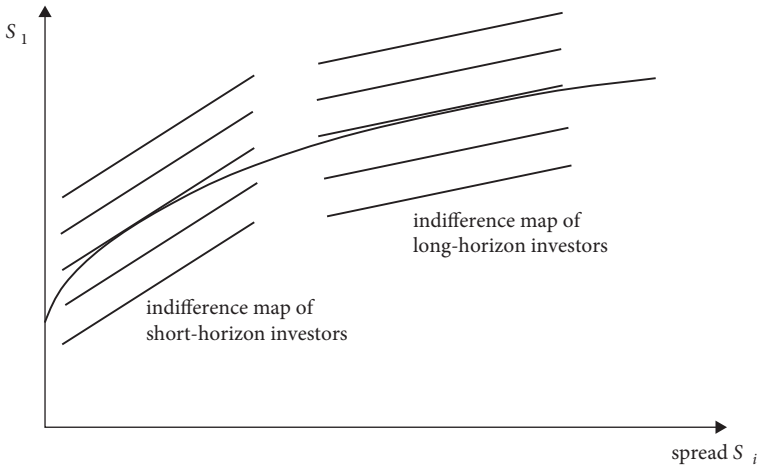


Figure 10.3 Expected return and spread with many clienteles.

Extending the logic to three or more securities ranked by liquidity, we would find that a “preferred habitat” equilibrium requires gross returns to form a (weakly) concave locus such as that drawn in Figure 10.3. Clienteles with different holding periods congregate at different points on the curve. Therefore, the gross return increases with the spread s_h , but at a decreasing rate.

10.2.3 Evidence

Equation (10.9) is that illiquid securities must provide investors with a higher expected return to compensate them for their higher transaction costs, controlling for the determinants of the required net rate of return: the security’s market risk, as measured by its beta. Observe that the realized gross return R is the expected return $E(R_j)$ plus an expectation error. If investors have rational expectations, this error will have zero mean and be independent of any other observed variable. Thus, according to equation (10.9), the observed gross return for security j should satisfy:

$$R_i = \gamma_0 + \gamma_1 \beta_i + \gamma_2 s_i + \varepsilon_i, \quad (10.11)$$

where γ_0 is an estimate of the risk-free return, γ_1 is an estimate of the risk premium on the market portfolio and γ_2 is an inverse measure of the average holding period.

Equation (10.11) can be tested and estimated by running a cross-sectional regression of stock returns on betas and relative bid-ask spreads. Such an estimation was first applied to monthly returns by Amihud and Mendelson (1986)

using NYSE and AMEX stock data for 1961–80.³ In their basic specification, which also includes year dummies, they obtain the following estimates for equation (10.11):

$$R_i = 0.0036 + 0.00672\beta_i + 0.211s_i,$$

(6.18) (6.83)

(t-statistics in parenthesis). These estimates imply that a 1-percentage-point increase in the bid-ask spread for a stock is associated with a 0.211% increase in its monthly expected return—that is, more than 2.5% per year. This coefficient also suggests that the marginal investor trades once every five months (holding period = $\frac{1}{0.211} \simeq 5$).

If—as is argued in Section 10.2.2—the relationship between expected return and spread is concave, it can be estimated as a piecewise linear regression. Amihud and Mendelson (1986) do this and find that the slope of the relationship does tend to decline as the spread increases. They also estimate other specifications, controlling for additional firm-specific variables that may affect required stock returns, such as firm size, tax treatment, and market-to-book ratio; they find that the effect of the spread is robust to such changes in specification and that it accounts for a good portion of the so-called small firm effect (small firms offering higher returns than larger firms with comparable risk).

Amihud and Mendelson point out that the increases in required returns called for by larger spreads translate into substantial reductions in asset valuations:

Consider an asset which yields \$1 per month and has a bid-ask spread of 3.2% (as in our high-spread portfolio group) and its proper opportunity cost of capital is 2% per month, yielding a value of \$50. If the spread is reduced to 0.486% (as in our low-spread portfolio group), our estimates imply that the value of the asset would increase to \$75.8. about a 50% increase, suggesting a strong incentive for the firm to invest in increasing the liquidity of the claims it issues. (p. 246)

This estimate makes it clear that corporate financial policies that enhance liquidity have a considerable payoff in terms of asset valuation. For instance, a company may reduce its cost of capital by listing on a more liquid stock exchange. Foerster and Karolyi (2000) provide evidence of a 44 basis points decline in intraday effective spreads for 52 Canadian companies listing in the United States, and Miller (1999) finds that the market reaction to the issuance of

³ They use 49 portfolios sorted according to the previous-year average bid-ask spread for the stocks in the portfolios and the previously estimated betas of the portfolios. They regress the monthly returns of these portfolios on their average bid-ask spreads and their betas in the previous year.

American Depository Receipt (ADR) is highest for firms that list on more liquid and better-known markets such as the NYSE and Nasdaq. We will return to the impact of liquidity on corporate policies in Chapter 10.

Amihud and Mendelson's study has spawned a vast empirical literature. Many subsequent studies confirmed the significant positive cross-sectional association between various illiquidity measures and asset returns, controlling for risk (e.g., Brennan and Subrahmanyam 1996 and Amihud 2002). Brennan and Subrahmanyam use the coefficient of price impact regressions (see Chapter 5) to develop a measure of illiquidity. If λ_i is the price impact coefficient for stock i , then its illiquidity in a given month is measured by $\lambda_i q_i / P_i$, where q_i is the average trade size in that month for stock i and P_i is its average price. Using NYSE stock returns for 1984–91, they also find a positive cross-sectional relationship between monthly average stock returns and this measure of illiquidity, after controlling for other factors that are known to affect stock returns (e.g., the so-called Fama-French factors).

One problem with this approach is that illiquidity may simply be proxying for some risk factor that has not yet been identified. One way to cope with this problem is to compare returns for securities that have similar risks but different illiquidity. Bonds issued by highly rated sovereign issuers are ideal candidates, since they present no risk of default. They should therefore offer the same yields if they have the same residual maturity. Amihud and Mendelson (1991) were first to use this approach. As the beginning of Section 10.2.1 explains, they show that the yield to maturity of U.S. Treasury notes with six months or less to maturity exceeds that on Treasury bills with similar maturity. This supports the theory since Treasury bills are more liquid. Their findings are also confirmed by subsequent studies (e.g., Kamara 1994 or Longstaff 2004).

Differential liquidity also provides a possible explanation for the “on/off-the-run” yield differential: “off-the-run” issues, like T-bonds and T-notes issued before the last auction for the same instruments, are typically less liquid than “on-the-run” issues for the same maturity. Thus, on-the-run bonds should offer lower yields. And this is indeed the case, as found empirically in various studies (Krishnamurthy 2002 or Goldreich, Hanke, and Nath 2005, among others).⁴ Goldreich et al. (2005) show that the yield differential between on-the-run and off-the-run securities narrows as the date at which securities go off-the-run nears. This finding shows that investors are forward looking and care about future rather than current liquidity in pricing on-the-run securities (this is

⁴ On-the-run bonds are often borrowed by traders making short sales (to play on the yield difference between these bond and the off-the-run bonds). Holders of on-the-run bonds can therefore earn lending fees from short sellers. These fees also help to widen the yield spread between on and off the run bonds (see Duffie et al. 2002 and Krishnamurthy 2002).

consistent with the fact the gross required return on a security depends both on current liquidity, s_t , and future liquidity, s_{t+h}).

Another way to assess the impact of liquidity on asset prices and the cost of capital is to consider changes in trading rules known to affect liquidity (see exercise 2). For instance, Muscarella and Piwowar (2001) study a sample of 134 stocks listed on the Paris Bourse that switched from call trading to continuous trading (86 stocks) or vice versa (48 stocks). Those switching from call to continuous trading improved their liquidity, while those going over to call trading suffered a deterioration. Accordingly, a switch from periodic call auctions to continuous trading should reduce the cost of capital and produce a jump in stock's price between the date at which the switch is announced and the date at which it is implemented. A switch from continuous trading to a call market should have the opposite effect. This is precisely what Muscarella and Piwowar (2001) find: stocks that switch to continuous trading register a price rise of about 5.5% while those switching to call trading undergo a decline of 4.9%.

10.3 Illiquidity Premia and Sources of the Spread

In the previous section, the bid-ask spread was taken to be exogenous. In Chapter 3 we have seen that the bid-ask spread in turn depends on asymmetric information, order-processing costs, inventory holding costs, and the rents arising from imperfect competition among market participants. Hence, ultimately each of these determinants of transaction costs should affect asset prices. And by the same token, they should affect asset returns via the illiquidity premium (Section 10.2.1). In this section we show that therefore the illiquidity premium is increasing in the likelihood of informed trading, order processing costs and inventory holding costs, and report evidence supporting some of these predictions. The next section will show that in the context of OTC markets, where trade is not centralized, the illiquidity premium is also affected by search frictions, both directly and via their effect on the bid-ask spread, and report some relevant evidence from the corporate bond market.

10.3.1 Predictions from Models of the Bid-Ask Spread

To illustrate the impact of determinants of the bid-ask spread on assets' illiquidity premia in the simplest possible way, we consider the expressions for the bid-ask spread derived in the one-period models presented in Chapter 3.

Recall that in the model where informed and liquidity traders place orders with uninformed, risk-neutral dealers, the opening bid-ask spread is

$S = \pi(v^H - v^L)$ (expression 3.12), and the midquote m equals the unconditional estimate μ of the fundamental value. Expressing the final value of the security as $v^H = \mu(1 + \sigma)$ when high and $v^L = \mu(1 - \sigma)$ when low, where σ measures the advantage of informed traders, the relative spread is:

$$s = \frac{S}{m} = \frac{\pi(v^H - v^L)}{\mu} = 2\pi\sigma.$$

Therefore, for small values of π and σ , using the approximation for the gross return in expression (10.6) for $h = 1$, the illiquidity premium becomes:

$$R - r \simeq 2\pi\sigma, \quad (10.12)$$

and therefore it is greater for securities where informed traders are more numerous and have more valuable information. This prediction also holds in a more general model of asymmetric information with multiple securities and risk averse investors. Easley and O'Hara (2004) consider such a model, where investors observe public signals about payoffs and a fraction of them also get private signals. In equilibrium, the expected return on a security is higher when the ratio of private to public signals is larger. In the presence of informed investors, the uninformed end up holding more securities with low payoffs and fewer securities with high payoffs than they would in a market without asymmetric information. Thus, they require a higher return on their holdings.⁵ Easley and O'Hara (2004) note that holding several securities does not alleviate this problem, since uninformed investors always end up on the wrong side of the trade.

As shown in Chapter 3, the model of the bid-ask spread based on adverse selection can be easily augmented with order-processing costs and/or imperfect competition rents. Assuming that these amount to γ per dollar traded, the bid-ask spread becomes:

$$s = 2(\pi\sigma + \gamma),$$

so that the illiquidity premium on the asset also reflects the market makers' order processing costs and oligopoly rents:

$$R - r \simeq 2(\pi\sigma + \gamma). \quad (10.13)$$

⁵ Another reason is that uninformed investors face greater uncertainty when the number of public signals decreases relative to private signals. As they are risk averse, they require a higher premium.

Finally, recall that in the one-period version of the inventory holding cost model presented in Chapter 3, according to expression (3.46), the relative bid-ask spread faced by an investor placing a unit-size order $d = \pm 1$ is $s \equiv S/\mu = 2\rho\sigma_\varepsilon^2/\mu$. Hence, using again expression (10.6) for $h = 1$, in this case the illiquidity premium is approximately:

$$R - r \simeq 2\rho\sigma_\varepsilon^2/\mu = 2\rho\mu\sigma_r^2, \quad (10.14)$$

where $\sigma_r \equiv \sigma_\varepsilon/\mu$ is the standard deviation of the percentage return on the security. Thus, the illiquidity premium on the asset reflects not only the volatility of fundamentals σ_r^2 but also the dealers' risk aversion ρ and the expected dollar value traded μ .

10.3.2 Evidence

Asymmetric Information. Easley, Hvidkjaer, and O'Hara (2002) test the prediction that stocks more exposed to informed trading command a higher illiquidity premium, using the probability of information-based trading (PIN) in a stock as a proxy for the level of informed trading.⁶ In Chapter 5, section 5.4, we explain how to estimate this probability using intraday data on buy and sell orders. Specifically, Easley et al. estimate the following regression in each month over the period between 1984 and 1998 for a large sample of stocks listed on the NYSE:

$$R_{it,l} = \gamma_{0t} + \gamma_{1t}\hat{\beta}_i + \gamma_{2t}PIN_{it-1} + \gamma_{3t}SIZE_{it-1} + \gamma_{4t}BM_{it-1} + \eta_{it}, \quad (10.15)$$

where $R_{it,l}$ is the excess return of stock i in month l of year t ; $\hat{\beta}_i$ is an estimate of the β for stock i over the entire estimation period; PIN_{it-1} is an estimate of the PIN measure using order data for stock i at the end of year $t - 1$; $SIZE_{it-1}$ is the logarithm of the stock's market capitalization of stock i at the end of year $t - 1$; BM_{it-1} is the logarithm of the stock's book-to-market value at the end of year $t - 1$. The authors control for market capitalization and book-to-market ratios because these variables are known to be priced characteristics (i.e., to explain the cross-section of stock returns).

⁶ Kelly and Ljungqvist (2012) use a different approach. They use a shock (a drop in the number of securities analysts covering a stock because of termination their research departments by brokerage houses) that increase asymmetric information in a subset of stocks (those covered by closed brokers). They show that the price of stock affected by this shock falls (the cost of capital increases).

Over all the monthly regressions, the average effect of PIN is positive and significant, in various specifications of equation (10.15). Thus, stocks with a higher likelihood of informed trading have higher returns on average, as predicted by equation (10.12). In principle, the PIN measure may capture other characteristics that affect returns. In particular, high-PIN stocks may also be illiquid, but for reasons other than just asymmetric information. However, the effect of PIN on returns does not vanish when the bid-ask spread is included as a control variable in equation (10.15). However, in a more recent study, Duarte and Young (2009) estimate equation (10.15) over a longer sample period (1984–2004) using the Amihud ratio (defined in Chapter 2) to control for illiquidity, and find that, once this variable is controlled for, the estimated coefficient of the PIN measure is no longer statistically significant.

Inventory Holding Costs. Friewald and Nagler (2019) study the effect of inventory holding costs on monthly bond yields (a measure of investors' required gross returns for investing in bonds), based on transaction prices for a large sample of U.S. corporate bonds from 2003 to 2013. Using these data, they estimate time series regressions of monthly changes in bond yields on aggregate factors that should affect these yields according to standard asset pricing models (such as changes in the 10-year Treasury rate, changes in the slope of the yield curve and changes in market volatility), as well as marketwide proxies of trading frictions specific to OTC markets, namely, inventory holding costs and search costs (that we shall analyze in Section 10.4).

They measure dealers' inventory holding costs in four ways: the change in their aggregate inventory in a given month, the change in the amount of bonds outstanding (as greater bond issuance in a given month presumably forces dealers to increase their inventories), and two proxies for dealers' risk aversion (parameter ρ in equation (10.14)), namely, the marketwide fraction of prearranged trades, for which dealers avoid inventory risk, and the TED spread, which in this context is used as a proxy of dealers' funding costs.⁷ In line with the prediction from equation (10.14), Friewald and Nagler (2019) find that bond yields are positively correlated with these measures of dealers' inventory holding costs, as well as search costs, while they are not significantly affected by asymmetric information, as the coefficient of the PIN measure is not significantly different from zero (though positive) in their estimates.

⁷ The TED spread is the difference between the three-month London Interbank Offered Rate (LIBOR) at which banks lend to each other and the three-month Treasury bill rate. As U.S. Treasury bills are seen as risk-free assets, the TED spread is commonly used as a measure of funding costs for dealers (see, for instance, Brunnermeier (2009) or Friewald and Nagler (2019)).

10.4 Illiquidity Premia in OTC Markets

As Chapters 1 and 8 explain, trading in OTC markets (such as swaps, CDSs, corporate bonds, etc.) is decentralized and opaque. As a result, it takes more time for investors in these markets to find a suitable counterparty and execute their trades. Moreover, opacity softens competition between dealers and therefore gives them bargaining power. In this section, we show how delays in execution affect illiquidity premia, both directly and indirectly via their effects on dealers' bid-ask spreads, as highlighted by a recent strand of research on asset pricing in OTC markets, pioneered by Duffie, Gârleanu, and Pedersen (2005, 2007).

10.4.1 Search Frictions and Illiquidity Premia

We consider a simplified version of the model by Duffie, Gârleanu, and Pedersen (2005), where a continuum of risk-neutral investors can invest in two different assets: a riskless security paying one dollar per period forever (a "consol bond") and a bank account offering a rate of interest r . The per-capita supply of the consol bond is denoted by q .

Investors can either hold one share of the security or none, and short-sales are not allowed. Hence, all trades are for a single share. Some investors have a holding cost of c per period: when they hold the asset, their cash flow is only $\$(1 - c)$ per period, and thus are "low-valuation" investors; instead, those who bear no holding cost are "high-valuation" investors. The holding cost of low-valuation investors may reflect, for instance, a less favorable tax rate than that faced by high-valuation investors. Thus, in each period t , there are four groups of investors: (i) high-valuation investors who own the security, (ii) high-valuation investors who do not own the security, (iii) low-valuation investors who own the security, and (iv) low-valuation investors who do not own the security.

In each period, after receiving their payoff, investors' valuation may switch (from low to high or high to low) with probability ψ . Intuitively, these shocks explain why investors want to trade in the model. Indeed, investors with a low valuation place a lower value on the asset than investors with a high valuation since they derive smaller cash-flows from holding it. Thus, there are gains from trade between investors with low valuations holding the asset and investors with high valuations not holding the asset. As soon as an investor's valuation switches from high to low, he wishes to sell the asset and starts looking for a potential buyer. For this reason, $1/\psi$ can be seen as investors' holding horizon (h in Section 10.1).

When their valuation switches, investors can trade the security. Investors willing to buy (high valuation non owners) or sell (low valuation owners) can

do so by trading with dealers at bid and ask prices b and a . However, finding a dealer takes time. Specifically, in each period, the probability of finding a market maker for an investor who wants to trade is ϕ , so that on average the execution delay for an investor is $1/\phi$ (as we shall see, ask and bid quotes are such that investors immediately trade when they find a counterparty). This delay is costly for investors who need to trade because it postpones the moment at which they obtain their preferred position in the asset. For instance, investors with a low valuation inefficiently bear the holding cost c as long as they hold the asset.

To sum up, in each period, the time line of the model is as follows:

1. Investors holding the asset receive their payoff, \$1 if they have a high valuation or $\$(1 - c)$ if they have a low valuation.
2. Each investor changes his valuation for the security with probability ψ .
3. High-valuation non-owners and low-valuation owners search for a dealer and find one with probability ϕ .

The fractions of investors' types and the prices quoted by dealers in each period are endogenous (the appendix of this chapter explains how they are determined). When the per capita supply of the asset is low (i.e., $q < 1/2$ as we assumed), the price paid by investors for the security turns out to be (see the appendix):

$$a = \frac{1}{r} - \underbrace{\frac{2\psi}{r(1+\chi)} \left(1 - \phi \frac{1-\chi}{2}\right)}_{\text{illiquidity premium}} S, \quad (10.16)$$

where S is dealers' bid-ask spread and χ is an index of dealers' market power when they buy the asset from sellers. This index is the fraction of the gains from trade captured by dealers when they trade with sellers: when $\chi = 1$, dealers' market power is maximal.

As explained in the appendix, a is the maximum price that investors with a high valuation are willing to pay for the asset.⁸ Indeed, this is the present (discounted) value of the cash-flows that they expect to receive from the asset, including the resale value when their valuation for the asset will become low. For instance, if $\psi = 0$, investors with a high valuation never need to resell the asset (their holding horizon is infinite). Thus, after buying the asset, they receive a stream of cash flows of \$1 per period forever, whose present value is $1/r$. When

⁸ Thus dealers extract all the surplus from trades with buyers. The reason is that when $q < 1/2$, there is an excess of buyers relative to sellers at any date (see the appendix). Dealers' ask price is in fact the price that would clear the market if trading were centralized.

$\psi > 0$, this present value is less because (i) at some point (on average in $1/\psi$ periods) investors start bearing holding costs and (ii) bear these costs as long as they do not find a counterparty. Moreover, when they eventually find one dealer, investors bear a bid-ask spread cost S . In anticipation of these costs, buyers require a compensation for holding the asset, captured by the second term in the previous equation. This illiquidity premium compensates investors for (i) holding costs due to delays in finding a counterparty when they resell the asset and (ii) bid-ask spread costs. While the second effect is identical to that highlighted in Section 10.2, the first is novel.

Holding the bid-ask spread constant, the illiquidity premium increases with the probability that trading needs arise, ψ , and decreases with the probability of finding a dealer, ϕ . Thus, it is inversely related to investors' expected horizon, $1/\psi$ (as in Section 10.2) and positively related to their expected delay in execution of their desired trade, $1/\phi$. This shows that delays in execution also affect illiquidity premia.

These parameters also affect the equilibrium bid-ask spread, whose expression is (as shown in the appendix):

$$S = a - b = \frac{(1 + \chi)c}{2(r + 2\psi) + (1 - 2\psi)\phi(1 - \chi)}. \quad (10.17)$$

Thus, dealers' bid-ask spread increases with their bargaining power, χ . For this reason, the illiquidity premium also increases in dealers' bargaining power, as can be seen more easily upon substituting expression (10.17) for S in expression (10.16) for the illiquidity premium.⁹

The bid-ask spread also increases when investors expect longer delays for the execution of their orders, that is when the probability ϕ of finding a dealer decreases. The reason is that gains from trade between dealers and investors are larger when investors expect larger delays in finding a counterparty. For instance, consider a seller who just found a dealer. If the seller turns down the dealer's offer, he expects to bear holding costs for longer time when ϕ is smaller. Thus, the surplus created by an immediate transaction is larger. As dealers' bid price is set to capture a fraction χ of this surplus, dealers' bid-ask spread increases. In sum, an increase in execution delays affects the illiquidity premium both directly (i.e., holding trading costs constant, as explained previously) and indirectly via the bid-ask spread.

⁹ The bid-ask spread does not vanish when χ goes to zero because dealers retain market power when they trade with buyers: this is because there is an excess of buyers relative to sellers when $q < \frac{1}{2}$, as explained in n. 6.

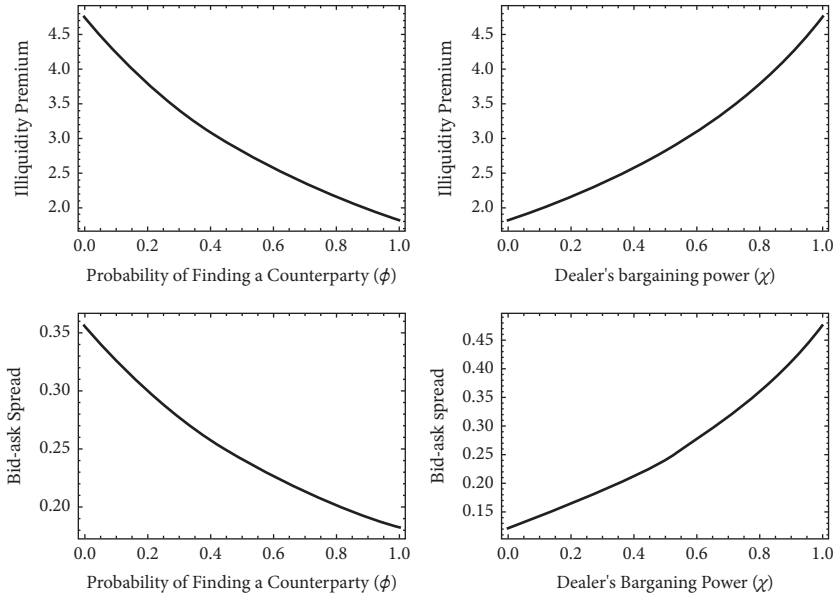


Figure 10.4 Effects of search and bargaining frictions on the illiquidity premium and the bid-ask spread.

Figure 10.4 illustrates the previous results for specific parameter values: $r = 1\%$ (so that the frictionless value of the asset is 100), $c = 0.1$ (i.e., holding costs equal 0.1% of the asset's frictionless value), $\psi = 0.1$ (implying that investors' holding horizon is 10 periods). The figures in the upper panel show the effect of ϕ (fixing dealers' bargaining power at $\chi = 0.5$), while those in the lower panel illustrate the effect of dealers' bargaining power, χ (for $\phi = 0.5$).

Overall, the results in this section provide a rationale for policies that induce a greater centralization of trading in OTC markets. Such policies should reduce illiquidity premia (and ultimately reduce the cost of capital for issuers) by reducing (i) execution delays due to search frictions and (ii) dealers' market power via more intense competition among dealers.

10.4.2 Evidence

As explained in Section 10.3.2, Friewald and Nagler (2019) provide evidence that inventory holding costs affect corporate bond yields, in line with equation (10.14). They also show that dealers' bargaining power (χ) and the time delay in trades due to search frictions ($1/\phi$) increase corporate bond yields. For instance, bond yields tend to rise in months in which trading is concentrated among fewer

dealers, so that they have greater bargaining power vis-à-vis their clients, as predicted by model presented above. Moreover, bond yields increase in months in which dealers appear less connected with each other (based on measures drawn from network analysis), so that delays in execution become larger, again in line with the model's predictions.

Overall, Friewald and Nagler (2019) conclude that inventory holding costs, search frictions and dealers' bargaining power jointly account for about 23% of the time-series variations in bond yields in their sample.

10.5 Liquidity Risk and Asset Prices

The previous sections established that future expected liquidity affects required rates of return. Now we turn to the effect of uncertainty about the future level of liquidity on the required rate of return. Intuitively, if liquidity fluctuates over time, it contributes to the volatility of returns: this source of risk is called liquidity risk. To see why and how unexpected changes in future liquidity can contribute to the risk of a security over and above its fundamental risk, we present two seminal approaches: the liquidity-adjusted version of the Capital Asset Pricing Model (LCAPM) developed by Acharya and Pedersen (2005) (Section 10.5.1), and the multifactor model proposed by Pastor and Stambaugh (2003) (Section 10.5.2).

10.5.1 The Liquidity-Adjusted CAPM

Suppose that all investors' holds securities for h periods, and consider again expression (10.5) for security j , denoting by $R_{j,t+h}$ the gross per-period rate of return of the security between date t and date $t+h$, and by $r_{j,t+h}$ its corresponding per-period net rate of return over the same holding horizon:

$$(1 + R_{j,t+h})^h = (1 + r_{j,t+h})^h \times \frac{1 + \frac{s_t}{2}}{1 - \frac{s_{t+h}}{2}}. \quad (10.18)$$

Taking a logarithmic approximation yields:

$$r_{j,t+h} \simeq R_{j,t+h} - \frac{s_{j,t+h} + s_{j,t}}{2h}. \quad (10.19)$$

Analogously the per-period net rate of return of the market portfolio is:

$$r_{M,t+h} \simeq R_{M,t+h} - \frac{s_{M,t+h} + s_{M,t}}{2h}, \quad (10.20)$$

where $s_{M,t}$ and $s_{M,t+h}$ are the average bid-ask spreads of the securities included in the market portfolio at dates t and $t+h$, respectively.

Recall that according to the CAPM, the risk premium for holding a security is proportional to the covariance of its return with the return of the market portfolio. More precisely, the net return required by investors is $E(r_{j,t+h}) = r + \beta_j [E(r_{M,t+h}) - r]$, which, using equations (10.19) and (10.20), can be rewritten as:

$$E(R_{j,t+h} - \frac{s_{j,t+h} + s_{j,t}}{2h}) = r + \beta_j [E(R_{M,t+h} - \frac{s_{M,t+h} + s_{M,t}}{2h}) - r]. \quad (10.21)$$

where security j 's *net* beta is defined as the covariance of security j 's *net* return with the *net* return of the market portfolio, divided by the variance of the net return on the market portfolio:

$$\beta_j \equiv \frac{\text{cov}(r_{j,t+h}, r_{M,t+h})}{\text{var}(r_{M,t+h})}. \quad (10.22)$$

Substituting in for the net returns from equations (10.19) and (10.20):

$$\beta_j = \frac{\text{cov}(R_{j,t+h} - \frac{s_{j,t+h}}{2h}, R_{M,t+h} - \frac{s_{M,t+h}}{2h})}{\text{var}(r_{M,t+h})}, \quad (10.23)$$

where we use the fact that at date t investors know the current value of the bid-ask spread of security j , $s_{j,t}$, and of the market portfolio, $s_{M,t}$, so that they do not affect covariances.

Thus the net beta of security j can be expressed as the sum of four betas:

$$\begin{aligned} \beta_j &= \frac{\text{var}(R_{M,t+h})}{\text{var}(r_{M,t+h})} \times \left[\frac{\text{cov}(R_{j,t+h}, R_{M,t+h})}{\text{var}(R_{M,t+h})} + \frac{1}{4h^2} \frac{\text{cov}(s_{j,t+h}, s_{M,t+h})}{\text{var}(R_{M,t+h})} \right. \\ &\quad \left. - \frac{1}{2h} \frac{\text{cov}(R_{j,t+h}, s_{M,t+h})}{\text{var}(R_{M,t+h})} - \frac{1}{2h} \frac{\text{cov}(s_{j,t+h}, R_{M,t+h})}{\text{var}(R_{M,t+h})} \right] \\ &= \frac{\text{var}(R_{M,t+h})}{\text{var}(r_{M,t+h})} \times \left[\beta_{j1} + \frac{1}{4h^2} \beta_{j2} - \frac{1}{2h} \beta_{j3} - \frac{1}{2h} \beta_{j4} \right]. \end{aligned} \quad (10.24)$$

Finally using this expression in equation (10.21), the required risk premium on security j can be written as:

$$E(R_{j,t+h}) - r = \beta_{j1} \lambda_M + \underbrace{\kappa(s_{j,t} + E(s_{j,t+h})) + \beta_{2j}(\kappa^2 \lambda_M) - \beta_{3j}(\kappa \lambda_M) - \beta_{4j}(\kappa \lambda_M)}_{\text{illiquidity premium}}, \quad (10.25)$$

where $\kappa \equiv 1/2h$ and λ_M is the adjusted risk premium on the market portfolio:

$$\lambda_M \equiv \frac{\text{var}(R_{M,t+h})}{\text{var}(r_{M,t+h})} \times [E(R_{M,t+h} - s_{M,t+h}) - r]. \quad (10.26)$$

The decomposition in equation (10.25) yields a liquidity-adjusted CAPM, which is relevant for any asset pricing test of the CAPM that relies on gross returns—that is, for tests relying on returns measured as percentage changes of midquote prices, rather than net returns. Equation (10.25) shows that the illiquidity premium includes two components: the first is the compensation for the current and expected level of trading costs, $s_{j,t} + E(s_{j,t+h})$, already discussed in Section 10.2.1 and captured in equation (10.9); the second component arises only if there is systematic illiquidity risk, as measured by the three illiquidity betas, β_{2j} , β_{3j} , and β_{4j} .

Each illiquidity beta captures a source of covariance risk arising from innovations in illiquidity: only β_{1j} captures the covariance between the security's fundamentals and those of the market portfolio; the other three are all related to illiquidity risk. Specifically:

1. Parameter $\beta_{2j} = \text{cov}(s_{j,t+h}, s_{M,t+h})/\text{var}(R_{M,t+h})$ measures the commonality in liquidity, namely, the extent to which the illiquidity of stock j ($s_{j,t+h}$) correlates with market-wide illiquidity ($s_{M,t+h}$). Holding stocks that remain liquid when others go illiquid is a way to hedge a drop in asset values due to market-wide illiquidity. Thus, other things being equal, these stocks are attractive and should have a higher price (lower risk premium). Conversely, stocks with a high β_2 must have greater returns on average, because investors require compensation for holding a security that becomes illiquid when the overall market does.
2. Parameter $\beta_{3j} = \text{cov}(R_{j,t+h}, s_{M,t+h})/\text{var}(R_{M,t+h})$ measures the sensitivity of individual security returns to market liquidity, namely, the co-movement between the gross return on stock j ($R_{j,t+h}$) and market-wide illiquidity ($s_{M,t+h}$). A high value of β_{3j} means that the stock does well when market-wide illiquidity increases. Thus, stocks with high β_{3j} offer a hedge against a drop in market-wide liquidity, so investors require a smaller risk premium to hold these stocks.
3. Parameter $\beta_{4j} = \text{cov}(s_{j,t+h}, R_{M,t+h})/\text{var}(R_{M,t+h})$ measures the sensitivity of individual securities' liquidity to market returns, namely, the co-movement between the illiquidity of stock j and the gross return on the market portfolio. Stocks with a high β_{4j} require a lower expected return, because they tend to remain liquid when the market is down: investors value such stocks more highly, as they allow them to sell at low transaction costs in an adverse market phase.

The model implies that, as any other type of risk, liquidity risk will be “priced” by financial markets only if it cannot be eliminated via diversification: if liquidity risk were purely idiosyncratic, security j ’s future bid-ask spread $s_{j,t+h}$ would by definition not comove either with market-wide illiquidity or with the return of the market portfolio, so that $\beta_{2j} = \beta_{4j} = 0$. Furthermore, a well-diversified portfolio of securities would feature a negligible amount of illiquidity risk, as the unexpected drop in the liquidity of some securities can be expected to be offset by unexpected increases in the liquidity of other securities in the portfolio. Hence, the liquidity risk of the market portfolio would be negligible, implying that $\beta_{3j} = 0$.

However, empirical studies indicate that liquidity risk is not idiosyncratic: Hasbrouck and Seppi (2001); Chordia, Roll, and Subrahmanyam (2000); and Huberman and Halka (2001) show that there is “commonality” in liquidity risk. That is, liquidity measures for various securities (e.g., bid-ask spreads or measures of price impact) are usually positively correlated. This co-movement in liquidity implies that the liquidity risk on a security cannot be easily diversified away and so contributes to its systematic risk. In this situation, the logic of asset pricing models (such as the CAPM) implies that investors will require compensation for liquidity risk, thereby creating a second link between the price and the liquidity of financial assets, beside that highlighted by Amihud and Mendelson (1986).

Acharya and Pedersen (2005) estimate the LCAPM relationship (10.26) on CRSP data for all common stocks listed on NYSE and AMEX from 1962–99 in two stages: first, they estimate the betas as defined above; second, they estimate equation (10.26) on the cross-section of returns and betas. They measure illiquidity by the Amihud ratio, i.e., the ratio between the absolute value of daily returns and the daily volume.¹⁰

Acharya and Pedersen report three main results. First, illiquid stocks (those with high average transaction costs) tend to be more exposed to illiquidity risk, as the absolute value of their liquidity betas (β_{2j} , β_{3j} , and β_{4j}) are larger. Second, the cross-section of stock returns is explained better by the liquidity-adjusted CAPM than by the traditional CAPM. Last, liquidity risk is priced: overall, the various sources of liquidity risk contribute a 1.1% difference in average annual returns between a portfolio of illiquid stocks and a portfolio of liquid stocks. This yield spread is largely due to the third source of illiquidity risk (captured by β_{4j}): investors seem to value especially highly those securities that remain liquid when the market is down. However, when expected illiquidity and the

¹⁰ Since the Amihud ratio is at best proportional to the transaction cost variable s considered by the model, reliance on this empirical measure of illiquidity may affect the estimate of the scaling factor κ in expression (10.25). As a result, the estimate of κ cannot be used to identify the investors’ holding period h .

liquidity betas are entered jointly with no restrictions in the specification, their impact on the risk premium is estimated imprecisely because of collinearity.

Both Acharya and Pedersen's (2005) original work and subsequent work, notably by Lee (2011) and Holden and Nam (2019), find similar "first-stage" results: most betas are significantly different from zero, and the first liquidity-related beta, β_{2j} , is positive for most stocks while the remaining two, β_{3j} and β_{4j} , are generally negative (though Holden and Nam find that the betas do not seem to be very stable over time when comparing pre- and post-2000 estimates). Thus for most stocks illiquidity co-moves with market-wide illiquidity, and it worsens when the general market is down, while the stock's return is negatively correlated with market-wide illiquidity.

Regarding the "second stage," which considers the impact of illiquidity on stock returns, the evidence is more mixed. All studies confirm the established finding that the average level of illiquidity, s_j , carries a significant premium. However, Holden and Nam's (2019) replication and extension do not consistently find estimated coefficients on the illiquidity β 's that are of the predicted sign and significantly different from zero. Moreover, the implication of the LCAPM, that all four β 's should be significant and carry an equal coefficient λ_M , is rejected. However, it should be pointed out that the hypothesis that these coefficients should all be equal is at odds with the predictions from equation (10.25). To see this, observe that in equation (10.25) the liquidity betas have different coefficients whenever the holding period h differs from $\frac{1}{2}$. A further reason to expect this hypothesis to fail is that the measure of illiquidity used is not the bid-ask spread, but at best proportional to it.¹¹

10.5.2 The Liquidity-Augmented Factor Model

Pastor and Stambaugh (2003) have proposed an alternative approach to test the pricing of liquidity risk, which rests on a measure of aggregate stock market liquidity as a risk factor (i.e., a source of risk for stock returns for which investors demand a risk premium). More specifically, they estimate the following regression for NYSE and AMEX stocks (over the 1962–2000 period):

$$r_{jt} = \beta_j^0 + \beta_1^L \mathcal{L}_t + \beta_j^M MKT_t + \beta_j^M SMB_t + \beta_j^H HML_t + \epsilon_{jt}, \quad (10.27)$$

¹¹ Acharya and Pedersen include the effect of κ in their discussion of the expected illiquidity premium but do not account for its importance in determining the relationship between the coefficients on the betas: their theory predicts that the regression coefficient of liquidity β_{2j} should be κ^2 times that of β_{1j} , while those of liquidity betas β_{3j} and β_{4j} should be $-\kappa$ times that of β_{1j} .

where r_{jt} is security j 's excess return (i.e., the difference between its rate of return and the risk-free interest rate), $\mathcal{L}_t$ is the Pastor and Stambaugh liquidity factor (defined below), MKT_t is the excess return on the market portfolio, SMB_t is the size factor (i.e., the difference between the return on a portfolio that is long on small-capitalization stocks and short on large-capitalization stocks), HML_t is the value factor (i.e., the difference between the return on a portfolio that is long on high-book-to-market stocks and short on high-book-to-market stocks). These last three factors were proposed by Fama and French (1993), and are widely used in empirical asset pricing research.¹²

The coefficient of the liquidity factor $\mathcal{L}_t$, $\beta_1^{\mathcal{L}}$, measures the extent to which the return of security j responds to changes in aggregate liquidity. It therefore captures a source of risk similar to the coefficient β_{3j} in equation (10.25) proposed by Acharya and Pedersen (2005). By construction the approach proposed by Pastor and Stambaugh (2003) focuses just on market-wide liquidity risk, but it controls for a larger set of potential risk factors, beside a security's covariance with the return on the market portfolio.

To construct their liquidity factor $\mathcal{L}_t$, Pastor and Stambaugh (2003) estimate monthly regressions of daily security-level stock returns (in excess of the return of the market portfolio) on a lagged measure of signed aggregate trading volume, which is intended to capture past order flow. These individual stocks' regressions produce a monthly estimate of the "reversal coefficient" for each security. These are aggregated across securities into a market-wide illiquidity time series at the monthly frequency, which is the liquidity factor $\mathcal{L}_t$. The idea is that order flow should trigger a price response that is followed by a reversal in the subsequent period, as predicted by the order-processing costs and inventory holding costs models presented in Chapter 3. Therefore, the stronger is this reversal, the more illiquid is a stock. Thus, $\mathcal{L}_t$ is negative (it captures reversals) and a lower value of $\mathcal{L}_t$ means an increase in market wide illiquidity.

The time series $\mathcal{L}_t$ features occasional downward spikes, corresponding to months with particularly low liquidity (large reversals). Most of these correspond to sharp market downturns, the largest of which occurred in the stock market crash of October 1987. In a subsequent extension to more recent data, Pastor and Stambaugh (2019) find that the time series of liquidity sharply drops at the time of the 2008 financial crisis. Moreover, this measure of liquidity tends to be low at times of high market volatility.

¹² The Fama-French factors are available on Kenneth French's website at: <https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/datalibrary.html>. The Pastor and Stambaugh (2003) liquidity factor is available on Lubos Pastor's website here: <https://faculty.chicagobooth.edu/lubos-pastor/data>.

The estimates of the loading of the liquidity factor (β_1^L) in equation (10.27) turn out to be positive and statistically significant, indicating that empirically returns tend to be low when illiquidity increases. Moreover, Pastor and Stambaugh (2003) show that there are systematic differences in the average returns of portfolios formed by stocks with different exposures to liquidity risk (β_1^L), after controlling for the exposure of these portfolios to the other factors included in equation (10.27). In particular, the differential in average returns, adjusted for exposure to other sources of risks in equation (10.27), of a portfolio of stocks with high loadings (β_1^L) on the liquidity factor and a portfolio of stocks with low loadings is 9.23 %. Thus, investors require a compensation for holding stocks that are more exposed to liquidity risks, after controlling for their exposure to other well known risk factors.

Li, Novi-Marx, and Velikov (2017) have replicated and extended out-of-sample the 2003 study, and have broadly confirmed its results. However, they have raised a debate about the relationship between the liquidity factor (β_1^L) and the momentum factor, which captures the tendency of stocks to maintain recent price trends in the future. Pastor and Stambaugh (2019) provide a discussion of this issue.

10.6 Further Reading

Amihud, Mendelson, and Pedersen (2005) and Vayanos and Wang (2009) offer in-depth examinations of the effects of illiquidity on asset prices (see also Cochrane 2005). The seminal study by Amihud and Mendelson (1986) was followed by a spate of works confirming the positive cross-sectional relationship between liquidity and returns, controlling for risk: see Brennan and Subrahmanyam (1996); Datar, Naik, and Radcliffe (1998); and Chordia, Roll, and Subrahmanyam (2000); Similarly, the liquidity effects found by Amihud and Mendelson (1991) for fixed-income securities were confirmed by other studies, such as Warga (1992), Daves and Ehrhardt (1993), Kamara (1994), and Krishnamurthy (2002). Some researchers have also considered the effect of exogenous transaction costs on asset prices in dynamic portfolio choices models (e.g., Constantinides 1986 and Vayanos 1998).

It is worth pointing out that asymmetric information poses several important issues for asset pricing beyond those discussed in Section 10.2.4. One issue is which are the effects of asymmetric information in dynamic models of trading. Wang (1993) considers such a model with a single security, and finds that the effect of asymmetric information on expected returns conditional on a liquidity shock is ambiguous. An increase in the fraction of informed traders in his setup has two opposite effects. It increases adverse selection risk for uninformed

investors, which works to increase expected returns, but it also increases the amount of information available to investors, which reduces the uncertainty over the final payoff and with it the risk premium required by investors to absorb liquidity shocks.

Another question is whether, in the presence of asymmetric information, a CAPM-like relationship still obtains. Even under the usual CAPM assumptions (i.e., CARA utility functions and normal distributions of securities' pay-offs), investors with heterogeneous information have different expectations for returns and their covariance structure. Hence, they do not have the same mean-variance frontier, implying that under asymmetric information the usual two-funds separation theorem of the CAPM does not hold. Biais, Bossaerts, and Spatt (2010) show however that a CAPM-like relationship can be obtained by considering the weighted-average beliefs of informed and uninformed investors (see also Admati 1985).

The model of asset pricing in OTC markets in Section 10.2.5 is based on Duffie, Gârleanu, and Pedersen (2005), from which our presentation differs in two respects. First, their model is cast in continuous time, while we consider a discrete time. Second, in their model investors can trade directly with other investors, as well as with dealers. This model has pioneered the so called search-theoretic approach to OTC markets and gave rise to many subsequent extensions and applications. For instance, Duffie, Gârleanu, and Pedersen (2007) consider the case of a risky security and risk-averse investors, and analyze the dynamics of prices after an aggregate liquidity shock (an exogenous drop in the fraction of investors with a high valuation). Weill (2007) allows market makers to carry inventories over time and thus to counter the negative price effects of liquidity shocks. Vayanos and Wang (2007) posit multiple securities and derive cross-sectional implications for asset returns. See Weill (2020) for a survey of the search-theoretic approach to model trading in OTC markets.

Pastor and Stambaugh (2003) and Acharya and Pedersen (2005) were among the first to study whether liquidity risk is priced. Their approach and findings generated a large following literature and debates about whether liquidity risk is actually priced: see Li et al. (2019), Pastor and Stambaugh (2019), Pontiff and Singla (2019) or Ben-Rephael et al. (2015).

In contrast to the cross-sectional evidence, the longitudinal evidence on the relationship between bid-ask spread and returns is somewhat inconclusive. Using a very long series of monthly data, Jones (2002) finds that at times when spreads are large and so expected to be high in the near future, then expected returns are also high. However, Hasbrouck (2005) gets mixed results using daily data. These conflicting findings may be partly produced by the difficulty of disentangling the impact of time-varying liquidity from that of time-varying volatility, induced perhaps by bouts of noise traders' buy or sell orders.

A few papers highlight the interactions between illiquidity and fundamental risk. For instance, in Vayanos (2004), fund managers are subject to withdrawals when their performance falls below a given threshold. Hence, they are more likely to liquidate when the market is volatile, which increases the illiquidity premium at times of high volatility. In contrast, Favero, Pagano, and von Thadden (2010) predict a negative relationship between the illiquidity premium and volatility, as in their model investors have less use for liquidity when volatility is high since outside investment opportunities also deteriorate.

10.7 Appendix: The Derivation of the Search Model

This appendix shows how to derive the expressions for the ask price and the bid-ask spread in Section 10.4. In each period t , there are four groups of investors in the market: (i) high-valuation investors who own the security, (ii) high-valuation investors who do not own the security, (iii) low-valuation investors who own the security, and (iv) low-valuation investors who do not own the security. Let π_h^o , π_h^{no} , π_l^o , and π_l^{no} be the fractions of investors in each group, with superscript h and l referring respectively to a high-valuation and a low-valuation individual, and subscripts o and no to an owner and a non-owner.

In each period, the fraction of investors willing to buy the security is $\pi_b = \psi\pi_l^{no} + (1 - \psi)\pi_h^{no}$, and the fraction willing to sell is $\pi_s = \psi\pi_h^o + (1 - \psi)\pi_l^o$. We first show that when $q < \frac{1}{2}$, we have $\pi_b > \pi_s$. That is, at any date, there are more investors who want to buy the asset than investors willing to sell it.

To see this, let π_h be the steady-state fraction of high-valuation and π_l be the steady-state fraction of low-valuation investors. It must be that $\pi_h + \pi_l = 1$. Next, remember that in each period a fraction ψ of high-valuation investors become low-valuation investors, and vice-versa. Hence:

$$\pi_h = (1 - \psi)\pi_h + \psi\pi_l = (1 - \psi)\pi_h + \psi(1 - \pi_h). \quad (10.28)$$

Solving this equation for π_h yields $\pi_h = \frac{1}{2}$. By definition, $\pi_h = \pi_h^o + \pi_h^{no}$ and $\pi_l = \pi_l^o + \pi_l^{no}$. Moreover, all shares are necessarily owned either by high-valuation investors or by low-valuation ones. Hence $q = \pi_l^o + \pi_l^h$. Thus, we have:

$$\pi_h^o + \pi_h^{no} = \frac{1}{2},$$

$$\pi_l^o + \pi_l^{no} = \frac{1}{2},$$

$$\pi_l^o + \pi_h^o = q.$$

Using these equations, after some straightforward steps one obtains:

$$\pi_b = \pi_s + \left(\frac{1}{2} - q\right),$$

so that $\pi_b > \pi_s$ if and only if $q < \frac{1}{2}$.

Some dealers are contacted by sellers while some are contacted by buyers. Thus, after trading with their clients, dealers are either short one share of the asset or long one share of the asset. Following Duffie et al. (2005), we assume that dealers cannot hold an inventory position at the end of each period. This assumption has two implications. First, dealers must trade together (those with a short position buy the asset from those with a long position). We denote by μ the price at which interdealer trades takes place. Second, dealers' aggregate position after trading with their clients must be zero. As there are more investors seeking to buy the asset than investors seeking to sell it ($\pi_b > \pi_s$), this no aggregate inventory condition implies that some buyers of the asset must eventually decide not buy it. Ultimately, this decision depends on how dealers' ask and bid prices are set, which we explain now.

First, consider the case of a dealer who is contacted by an investor willing to sell the asset and let $\bar{b}$ be the minimum price that this investor is ready to accept. In this case, the potential gains from trade (surplus) between the investor and the dealer are $\mu - \bar{b}$. Following, Duffie et al. (2005), we assume that dealers set their bid price to obtain a fraction χ of this surplus. Thus, dealers' bid price is:

$$b = \chi \bar{b} + (1 - \chi)\mu,$$

with $0 \leq \chi \leq 1$. As dealers' bargaining power χ increases, they extract a larger surplus from sellers. At the limit, for $\chi = 1$, the surplus left to sellers is zero. As $b \geq \bar{b}$, an investor willing to sell the asset trades as soon as he finds a dealer. Thus, there is a mass $\phi\pi^s$ of dealers with a long position after trading with investors in each period.

Now consider the case of a dealer who is contacted by an investor willing to buy the asset and let $\bar{a}$ be the maximum price that this investor is ready to pay. At any ask price a strictly less than $\bar{a}$, the investor is strictly better off doing so and at $a = \bar{a}$ he is indifferent between trading or not. Thus, if $a < \bar{a}$, the mass of dealers with a short position after trading with investors is $\phi\pi^b$ and therefore exceeds the mass of dealers with a long position, $\phi\pi^s$ (since $\pi^b > \pi^s$). In this case, the zero aggregate inventory condition cannot be satisfied. In contrast, at $a = \bar{a}$, one can assume that only a fraction $\frac{\pi^s}{\pi^b}$ of investors buy the asset because investors willing to do so are indifferent between buying or not. In this way, the mass

of dealers with a short position after trading with investors is $\frac{\pi_s}{\pi_b} \times \phi \pi^b = \phi \pi^s$ and in aggregate dealers' position is zero, as required. In sum, this means that $a = \bar{a}$ and the likelihood that an investor willing to buy the asset trades in a given period is $\rho^b \equiv \phi \cdot \frac{\pi_s}{\pi_b}$.¹³

Finally, following Duffie et al. (2005), assume that these inter-dealer transactions take place at the midpoint between investors' reservations prices:

$$\mu = \frac{\bar{a} + \bar{b}}{2}. \quad (10.29)$$

To obtain the equilibrium ask and bid prices, we must now compute $\bar{a}$ and $\bar{b}$. To this end, we first compute the discounted value of the future stream of cash flows that each type of investor expects to receive just after trading in a given period (i.e., just after stage 3). Let V_j^k be this discounted value for a trader with valuation $j \in \{h, l\}$ and type $k \in \{o, no\}$.

Now, consider a high-valuation non-owner contacting a dealer: he buys the security if and only if:

$$V_h^o - a \geq V_h^{no},$$

since otherwise he is better off staying a non-owner. Similarly, a low-valuation owner will sell the security to a market maker if and only if:

$$V_l^{no} + b \geq V_l^o.$$

Thus, $\bar{a} = V_h^o - V_h^{no} = \Delta V_h$ and $\bar{b} = V_l^o - V_l^{no} = \Delta V_l$. To determine ΔV_h and ΔV_l , we first calculate V_j^k for $j \in \{h, l\}$ and $k \in \{o, no\}$. The value placed by a high-valuation owner on the security is:

$$V_h^o = \frac{1}{1+r} + \frac{(1-\psi)V_h^o}{1+r} + \frac{\psi(1-\phi)V_l^o}{1+r} + \frac{\psi\phi(V_l^{no} + b)}{1+r}. \quad (10.30)$$

To understand this expression, observe that a high-valuation owner always receives \$1 with certainty at the beginning of the next period, which explains the first term here. The last three terms are simply the weighted average of

¹³ Note that $a = \bar{a}$ means that dealers extract all surplus when they trade with a buyer while they don't when they trade with a seller. This makes sense because there is an excess of buyers relative to sellers. In fact, $\bar{a}$ is the price that would be set to clear the market if trading was centralized as in a Walrassian market.

the discounted cash flow for the investor in each of his possible states at the end of the next period, the weights being the respective probabilities. With probability $1 - \psi$, the investor remains a high-valuation owner and therefore values the discounted cash flow of the asset at V_h^o . This explains the second term in the equation. With probability $\psi(1 - \phi)$, he turns into a low-valuation owner who does not manage to sell to a dealer, and thus the subsequent period ends up valuing it at V_l^o ; this explains the third term. Finally, with probability $\psi\phi$, the investor becomes a low-valuation investor who does manage to resell the security at price b in the next period. In this state, the investor receives b but also keeps the option of buying at some point in the future. The value of this option is V_l^{no} . This state is captured by the last term.

Proceeding in the same way, we obtain the discounted value of future cash flows for a high-valuation investor who does not own the asset yet:

$$V_h^{no} = \frac{\psi V_l^{no}}{1+r} + \frac{(1-\psi)(1-\rho^b)V_h^{no}}{1+r} + \frac{(1-\psi)\rho^b(V_h^o - \bar{a})}{1+r}. \quad (10.31)$$

The first term here corresponds to the state in which the investor's valuation drops, so that in the next period he does not want to buy the asset anymore; the second to the state in which his valuation stays high but he does not manage to buy the security from a dealer; and the last term to the situation in which the investor does buy the asset from a dealer (at price $\bar{a}$ with probability ρ^b) and therefore owns the asset at the end of the next period. Following the same reasoning, we get:

$$V_l^o = \frac{1-c}{1+r} + \frac{\psi V_h^o}{1+r} + \frac{(1-\psi)(1-\phi)V_l^o}{1+r} + \frac{(1-\psi)\phi(V_l^{no} + b)}{1+r} \quad (10.32)$$

$$V_l^{no} = \frac{(1-\psi)V_l^{no}}{1+r} + \frac{\psi(1-\rho^b)V_h^{no}}{1+r} + \frac{\psi\rho^b(V_h^o - \bar{a})}{1+r} \quad (10.33)$$

Remember that $\Delta V_h = V_h^o - V_h^{no}$. From equations (10.30), (10.31), (10.32), and (10.33), we obtain:

$$\Delta V_h = \frac{(1-\psi(1-\phi))\Delta V_l + (\psi b + (1-\psi)\bar{a})\phi - (1-\psi)(\phi - \rho^b)(\Delta V_h - \bar{a})}{(1+r) - (1-\psi)(1-\phi)}.$$

Recalling that $\bar{a} = \Delta V_h$, this expression can be rewritten as:

$$\Delta V_h = \frac{1 + \psi(1-\phi)\Delta V_l + (\psi b + (1-\psi)\bar{a})\phi}{(1+r) - (1-\psi)(1-\phi)}. \quad (10.34)$$

Proceeding in the same way and remembering that $\Delta V_l = V_l^{no} - V_l^o$, we obtain:

$$\Delta V_l = \frac{(1-c) + \psi(1-\phi)\Delta V_h + (1-\psi)\phi b + \psi\phi\bar{a}}{(1+r) - (1-\psi)(1-\phi)}. \quad (10.35)$$

Hence:

$$\Delta V_h - \Delta V_l = \frac{c + S(1-2\psi)\phi}{(1+r) - (1-2\psi)(1-\phi)}, \quad (10.36)$$

where $S = a - b$ is the bid-ask spread charged by dealers. Now, recalling that:

$$\bar{a} = \Delta V_h, \quad (10.37)$$

$$b = \chi\bar{b} + (1-\chi)\mu. \quad (10.38)$$

and that $\mu = \frac{\bar{a} + \bar{b}}{2} = \frac{\Delta V_h + \Delta V_l}{2}$, we deduce from equations (10.37) and (10.38) that:

$$\Delta V_h - \Delta V_l = \frac{2(\bar{a} - b)}{1 + \chi} = \frac{2S}{1 + \chi}. \quad (10.39)$$

Substituting $\Delta V_h - \Delta V_l$ from this expression into equation (10.36) and solving for S , we get:

$$S = \frac{(1+\chi)c}{2(r+2\psi) + (1-2\psi)\phi(1-\chi)}, \quad (10.40)$$

which is expression (10.17) in Section 10.4.1.

Next, using the fact that $\Delta V_l = \Delta V_h - \frac{2S}{1+\chi}$ from equation (10.39) and $\bar{a} = \Delta V_h$, we can solve equation (10.34) for ΔV_h and obtain the ask price:

$$a = \bar{a} = \Delta V_h = \frac{1}{r} - \frac{2\psi}{r(1+\chi)} \left(1 - \phi \frac{1-\chi}{2} \right) S,$$

which is expression (10.16) in Section 10.4.1.

10.8 Exercises

- 1. Liquidity premium in the presence of dividend income.** Assume that investors hold a stock for one period, and can trade it at a constant percentage bid-ask spread s . The stock's fundamental value at date t , μ_t , equals its midprice at that date. The stock yields a constant dividend yield

d per period, so that the dividend per share at date t is $d\mu_t$. The one-period required rate of return on the stock is given and equal to r .

- a. Define the gross-of-transaction-cost return $1 + R$ in terms of μ_t , μ_{t+1} , and d .
 - b. Determine the equilibrium gross return $1 + R$ as a function of r , s , and d alone.
 - c. How does the liquidity premium $R - r$ change with the dividend yield d ? [Hint: recall that $(1 + x)/(1 + y) \approx x - y$, for x and y small.]
 - d. Explain the intuitive reason for the result obtained in answering point c.
2. **Transaction costs and clientele effects.** Consider two bonds: bond A yields 2% and has a bid-ask spread of 1.00%, while bond B yields 2.40% and has a bid-ask spread of 2%. Your asset management company caters to two groups of investors: the members of group 1 have a four-year expected holding period, while group 2 has a two-year expected holding period. The company asks you which of the two bonds should be marketed to which group of customers.
- a. Does group 1 prefer bond A or B ?
 - b. Does group 2 prefer bond A or B ?
 - c. Denote by $\Delta R = R_B - R_A$ the difference between the gross yields of the two bonds. Given the holding periods of the two groups and the bid-ask spreads of the two bonds assumed above, which are the minimum and maximum yield differentials ΔR that would produce the same clientele equilibrium as in the case considered under **a** and **b**?
 - d. Suppose that ΔR were higher than the upper bound found under c: which bonds would the two groups want to buy?
 - e. Suppose that ΔR were lower than the lower bound found under c: which bonds would the two groups want to buy?
3. **Effect of changes in market structure on expected returns.** Consider an investor who plans to buy stock X at date t and resell it, after one year, at date $t + 1$. The required annual rate of return on this stock is r and each year the stock pays a dividend equal to $\bar{D}$ on average. Let b_{t+1} be the expected resale price at $t + 1$. As the market for stock X is illiquid, the resale price is lower than the expected fundamental value μ_{t+1} . Specifically, $b_{t+1} = \mu_{t+1}(1 - s/2)$ where s is the bid-ask spread. Let $a_t = \mu_t(1 + s/2)$ be the ask price that the investor is willing to buy at date t .
- a. Show that:

$$\mu_t = \frac{\bar{D}}{\left[(1 + r)\frac{1+s/2}{1-s/2} - 1\right]\left(1 - \frac{s}{2}\right)},$$

where r is the net expected return on the stock.

- b. Let $r_t = \frac{D_t + b_{t+1}}{a_t} - 1$ be the actual return of the stock over the period $[t, t + 1]$. Note that this differs from the net expected return simply because the actual dividend may differ from the expected dividend in every period. Of course, $E(D_t) = \bar{D}$. Let $r_t - r$ be the “abnormal” return of the stock over the period $[t, t + 1]$. What is the average abnormal net return?
- c. Now suppose that at some date τ , the stock exchange on which stock X is listed introduces a new trading system. Following this change, the bid-ask spread on stock X becomes s^* . What is the effect of this change in trading organization on the price of the stock at date τ ? If this change is unexpected until date τ , what is the effect on the expected abnormal return of the stock from date $\tau - 1$ to date τ .
- d. Suppose you measure the average gross and net annual returns of stock X over ten years before the change at date τ , excluding the year preceding date τ ; then you perform the same calculation with ten years of data after the change in the trading system, and finally compare the two figures. If the change in the trading system makes the market more liquid (i.e., $s^* < s$), what should be the outcome of this comparison?
- 4. Effect on the bid-ask spread of the probability of finding a dealer.** Consider the equilibrium bid-ask S in expression (10.17) in the search model of Section 10.4.
- a. Assuming $\chi < 1$, how does the bid-ask spread respond to changes in the probability ϕ of finding a dealer? How does the answer depend on the value of the probability ψ that the investor’s valuation will change in the future?
- b. What is the intuitive explanation for this result?
- 5. Bid-ask spreads and trade size in OTC markets.** The Excel file Ch10_ex5_data.xlsx, available on the companion website for the book, contains trades from the over-the-counter market for a Danish residential mortgage bond with a 1% coupon maturing in 2053.¹⁴ The data covers 698 trades from March 2021 to October 2021.
- a. For each trade, compute the quoted bid-ask half spread as a percentage of the midprice. Compare the magnitude of the average spread to the bid-ask spreads from equity markets reported in Table 2.2.
- b. Inspect sell orders with trade ID 15 and 16. Although they were executed almost at the same time, do they have a similar spread? Would such a difference in spreads be possible in a market with a centralized limit order book? What is different about over-the-counter markets?

¹⁴ Source: nasdaqomxnordic.com.

- c. Estimate a regression of the spread on the trade size. A priori, based on the Kyle model presented in Section 4.2.2, what sign would you expect the coefficient on trade size to have? What sign does the coefficient actually have?
- d. Consider the Duffie, Gârleanu, Pedersen (DGP) model of an over-the-counter market from Section 10.4. Suppose that there are two (identifiable) types of investors: sophisticated investors (such as hedge funds) and unsophisticated investors (such as retail investors). If sophisticated investors have higher bargaining power than unsophisticated investors and also trade larger amounts, can the DGP model explain the empirical finding from question b?
- e. Classify all orders as being submitted by sophisticated or unsophisticated investors by using the following crude rule: if the order size is a multiple of 10 million, the investor is sophisticated, if not the investor is unsophisticated. Now estimate a regression of the spread on the order size and a dummy variable indicating whether the investor is sophisticated (without an interaction term). Are the results consistent with the theoretical explanation proposed in question c?